frontiers in Education

TYPE Systematic Review
PUBLISHED 25 May 2026
DOI 10.3389/feduc.2026.1819092

OPEN ACCESS

EDITED BY
Noble Lo,
Lancaster University, United Kingdom

REVIEWED BY
Ana Dolores Vargas Sánchez,
Universidad de La Sabana, Colombia
Sandra Fernando,
Ravensbourne University London, United Kingdom
Stepan Hubalovsky,
University of Hradec Kralove, Czechia

*CORRESPONDENCE
Fatima Al Amri
✉ fatimaalamri110@gmail.com

RECEIVED 27 February 2026
REVISED 07 April 2026
ACCEPTED 15 April 2026
PUBLISHED 25 May 2026

CITATION
Al Amri F, Alias N and DeWitt D (2026) Systematic literature review: gamification in teacher professional training (2019–2025).
Front. Educ. 11:1819092.
doi: 10.3389/feduc.2026.1819092

Systematic literature review: gamification in teacher professional training (2019–2025)

Fatima Al Amri*, Norlidah Alias and Dorothy DeWitt

Curriculum and Instructional Technology Department, Faculty of Education, Universiti Malaya, Kuala Lumpur, Malaysia

Introduction: Gamification has emerged as a prominent strategy for enhancing teacher professional development. Despite growing interest, no systematic review has specifically examined gamification applications in teacher professional training contexts.

Methods: This systematic review followed PRISMA 2020 guidelines to examine gamification integration in teacher professional training. A comprehensive search across four databases (Scopus, Web of Science, PubMed, and Google Scholar) identified 1,776 records, of which 27 empirical studies from 12 countries (2019–2025) met the inclusion criteria.

Results: Achievement-oriented gamification elements dominated implementations: points systems (88.9%), badges and levels (48.1% each), and leaderboards (40.7%). Implementation platforms included multi-tool platforms (40.7%), quiz-based systems (37%), and blended approaches (28%). Methodological diversity emerged through experimental designs (29.6%), qualitative investigations (22.2%), and mixed methods (18.5%). All included studies reported positive outcomes across motivation, engagement, and digital competence domains. However, geographic concentration in Europe (74.1%), particularly Spain (51.9%), restricts generalizability.

Discussion: While findings support gamification as a viable strategy for teacher professional development, results should be interpreted cautiously given the likelihood of publication bias and predominant reliance on self-report measures. Critical gaps include the absence of longitudinal studies and limited theoretical integration. Future research should prioritize cross-cultural implementations and longitudinal designs to establish globally applicable strategies.

KEYWORDS
gamification, professional development, professional training, teacher learning, teachers, vocational training

1 Introduction

"Gamification" gained recognition in 2010 with major digital developments. Rapid advancement of computers and internet enabled multimedia-rich digital games, enhancing educational capabilities (Al-Shazly, 2019). Since 2015, gamification emerged in educational research as a fundamental teaching strategy, developing theoretical and practical foundations. By 2017, studies explored gamification as a motivational tool in e-learning (Sailer et al., 2017), emphasizing game elements in non-game contexts to enhance interaction and motivation (Deterding et al., 2011).

Teachers in the digital age face various challenges that require them to develop their skills and teaching methods, necessitating a fundamental shift in their traditional roles and the development of innovative curricula that keep pace with ongoing technological advancements. Today's teachers must connect educational technology with digital learning environments while acquiring the skills needed to adapt to continuous professional developments. This requires a deep understanding of how to effectively and impactfully integrate modern technologies into educational practices.

Gamification represents a significant technological innovation in educational research. This growing interest necessitates comprehensive review of literature on gamification in teacher professional training. Previous reviews by Dahlan et al. (2024) on employee training (2018-2022) and Mahat et al. (2022) on general education applications (2015-2022) did not sufficiently focus on teacher professional training applications, justifying this specialized study.

1.1 Theoretical foundations of gamification in teacher professional development

Gamification has been defined from multiple scholarly perspectives. Deterding et al. (2011) defined it as the use of game design elements in contexts other than play, in order to improve learners' experiences and increase their participation and effectiveness. Kapp (2012, p. 10) described it as using game mechanics, aesthetic values, and game-related ideas to improve levels of participation and motivation, enhance learners' learning processes, and enable them to solve problems. Landers (2014, p. 752) further characterised gamification as the use of play characteristics outside of their original context with the aim of influencing learners' behaviours and attitudes. Collectively, these definitions converge on a shared understanding: gamification applies game elements to non-game educational settings to drive motivation, engagement, and learning outcomes.

It is important to distinguish gamification from related concepts. Gamification is neither a game nor a play-based learning strategy; rather, it integrates game design elements into non-game contexts to motivate learners (Deterding et al., 2011). As Kapp (2012) clarifies, gamification uses game mechanics to improve participation and motivation without replacing the learning activity itself with a game, while Landers (2014) emphasises that its purpose is to influence behaviours and attitudes rather than to create game experiences. In this review, studies employing complete game environments or serious games were excluded, while studies integrating discrete game mechanics into professional training contexts were retained. Although the search strings included the term "game-based learning" to maximise retrieval, all included studies were verified to involve gamification elements rather than complete game-based interventions. Within gamification itself, Nicholson (2015) draws a further distinction between structural gamification which overlays game elements such as points and badges onto existing content as an extrinsic motivational layer and content-based gamification, which embeds game mechanics directly into the instructional design. This distinction is analytically significant for the present review, as the findings reveal a predominant reliance on structural mechanics across the identified studies.

Understanding how gamification functions in teacher professional development requires grounding in established learning theories that explain its mechanisms and effects. Four complementary theoretical perspectives inform this review.

Self-Determination Theory (SDT), proposed by Deci and Ryan (1985), identifies autonomy, competence, and relatedness as fundamental psychological needs driving sustained motivation. Gamification elements such as progressive levels and personalised feedback address competence needs, collaborative challenges satisfy relatedness, and flexible platform choices support autonomy collectively explaining why gamified professional training sustains engagement beyond initial novelty.

Flow Theory (Csikszentmihalyi, 1990) describes optimal engagement as a state achieved when challenge and skill are balanced. Level progression and adaptive difficulty systems in gamified training are designed to maintain this balance, preventing boredom or anxiety and sustaining productive professional learning.

Behaviourist and Reinforcement Theory, rooted in Skinner's (1938) operant conditioning framework, explains how points, badges, and reward systems function as positive reinforcers that strengthen desired learning behaviours through immediate feedback a mechanism directly reflected in the near-universal adoption of points systems identified across the reviewed studies.

Social Cognitive Theory (Bandura, 1986) highlights observational learning, self-efficacy, and social modelling as drivers of skill acquisition. Leaderboards and peer collaboration features activate these mechanisms by providing visible performance benchmarks and fostering vicarious learning, while badge credentialing systems strengthen professional self-efficacy through recognised achievement.

At the intersection of gaming and pedagogy, Lo and Mok (2019) argue that gaming literacy encompassing paratextual practices such as peer-authored guides, online forums, and community generated tutorials functions as a sociocultural scaffold that supports learner autonomy and community identification. These processes are directly relevant to gamified professional development contexts, where paratextual resources can facilitate teachers' navigation of game-like learning environments, encourage self-directed exploration, and strengthen identification with professional learning communities, extending the motivational explanations offered by SDT and Social Cognitive Theory by illuminating how learning actually unfolds within gamified ecologies.

Empirical and theoretical reviews have further substantiated these foundations. Seaborn and Fels (2015) conducted a comprehensive survey of gamification in theory and action, confirming its broad applicability across educational and professional contexts. At the element level, Mekler et al. (2017) demonstrated experimentally that individual gamification components including points, badges, and leaderboards exert distinct and measurable effects on intrinsic motivation and performance, a finding directly relevant to interpreting the diversity of element adoption patterns identified across the studies reviewed here.

Situating gamification within established professional development frameworks further enriches the analytical lens of this review. Desimone (2009) identified five core features of effective professional development content focus, active learning, coherence, duration, and collective participation against which gamified training designs can be evaluated. Guskey's (2002) five-level model of professional development evaluation, encompassing participant reactions, learning, organisational support, use of new knowledge, and student learning outcomes, provides a structured framework for interpreting the outcome data reported across the reviewed studies. Furthermore, the Technological Pedagogical Content Knowledge framework (TPACK; Mishra and Koehler, 2006) offers a lens for understanding how gamified platforms may simultaneously develop teachers' technological, pedagogical, and content knowledge helping explain the preponderance of achievement mechanics and digital platform adoption identified in the evidence base.

These five theoretical perspectives collectively explain how gamification motivates (SDT), engages (Flow), reinforces (Behaviourism), develops professional identity (Social Cognitive Theory), and scaffolds community-based learning (Gaming Literacy) forming the analytical lens guiding interpretation of gamification elements, platforms, and outcomes throughout this review.

1.2 Research questions

This study conducts a comprehensive systematic review of gamification literature in teacher professional training, examining application strategies and effectiveness in improving professional performance. The central research question is: How has gamification been used in teacher professional training? Sub-questions identify the most frequently used gamification elements, explore preferred training platforms, and analyse positive and negative outcomes emerging from these studies.

This systematic review critically analyses previous research on gamification in teacher professional training. The methodology section details study selection and analysis, followed by results and discussion with practical applications for future research directions.

The primary purpose of this study is to answer the central research question:

How has gamification been utilized in teachers' professional training?

From this question, several sub-questions arise:

1. What is the volume and temporal distribution of publications on gamification in teacher training?
2. What are the most common research methods and designs used in studies related to gamification in teacher training?
3. What are the demographic characteristics of participants in the studies, including their experience levels and educational contexts?
4. What are the core gamification elements and mechanisms implemented in the studies?
5. What types of platforms are used for gamification, and how are they integrated into existing educational infrastructures?
6. What are the outcomes related to motivation, engagement, and skill development associated with the use of gamification in teacher training?
7. What are the technical and cultural challenges faced in implementing gamification in teacher training?

Addressing these questions necessitates a critical review of previous research related to gamification in teachers' professional training, selected according to specific criteria. The subsequent section details the methodology employed, followed by a presentation of the results of the reviewed studies in relation to the research questions. Finally, the discussion section offers practical applications and suggestions for future research directions.

2 Method

This section details the methodology employed to gather studies related to gamified professional training.

Publication standards are essential for providing researchers with guidelines to conduct thorough and rigorous reviews. The PRISMA 2020 framework serves as an updated standard for systematic literature reviews (Page et al., 2021). Accordingly, this study was conducted following the PRISMA 2020 checklist (see Figure 1).

Systematic research strategies

The systematic research strategy in this study comprises three primary processes: identification, screening, and eligibility.

2.1 Identification

The identification process involves discovering keywords and terminology relevant to the study, specifically focusing on gamification and vocational training. Keywords are developed based on the research questions (Okoli, 2015) to enhance the search for comprehensive articles across selected databases. This study's identification process utilized keywords derived from previous research, which were subsequently expanded through various search techniques, including phrase searches, ampersands, wildcards, and domain function symbols. The complete search strings are presented in Table 1. It should be noted that Google Scholar does not support truncation operators; therefore, key variants including "gamification", "gamified", and "gamify" were searched separately within this database to ensure comprehensive retrieval.

The search was conducted between February 2024 and November 2025 across all four databases simultaneously to ensure temporal consistency and comprehensive coverage of retrieved records. The complete search strategies for all databases are reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Search extension (PRISMA-S; Rethlefsen et al., 2021), as detailed in Table 1. The completed PRISMA 2020 checklist is provided as a Supplementary File.

The term "gamification*" was selected to include articles discussing gamification, vocational training, e-training, and teacher training. Four primary databases were utilized: Scopus (for widespread systematic review use and extensive coverage), Web of Science (for academic research prominence), PubMed, and Google Scholar (for comprehensive resource availability and accessibility). While domain-specific databases such as ERIC, PsycINFO, and IEEE Xplore were not searched, the four selected databases provided broad and overlapping coverage of the peer-

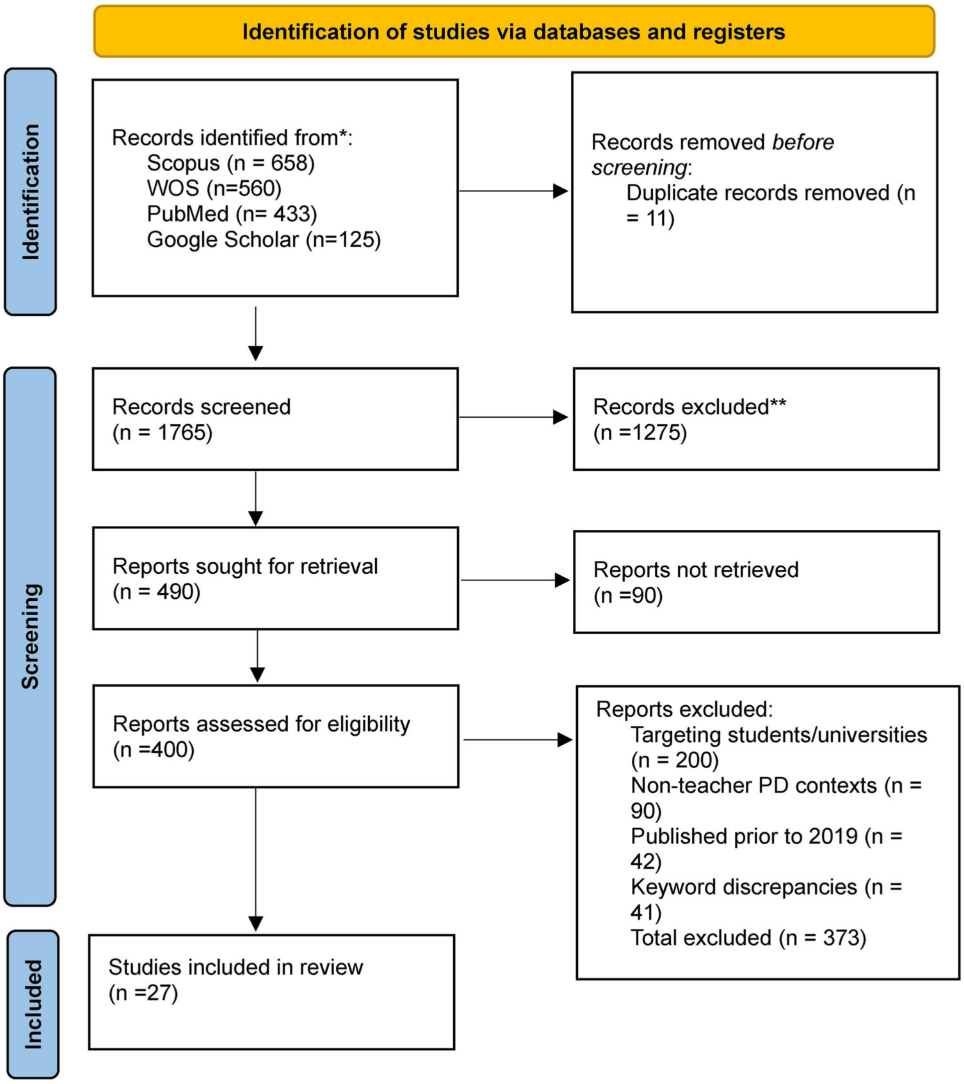

FIGURE 1
PRISMA 2020 flow diagram for systematic reviews using a template from Page et al. (2021) Adapted from the template by Page et al. (2021).

reviewed literature relevant to gamification in teacher professional training. In the initial phase, 1,776 articles were identified.

2.2 Screening

A total of 1,776 studies were initially identified across all four databases. After removing duplicates ($n = 11$), a total of 1,765 records were screened based on titles and abstracts, of which 1,275 were excluded as they did not meet the inclusion criteria. The remaining 490 studies were sought for full-text retrieval; however, 90 reports could not be retrieved. Retrieval was attempted through institutional library access and direct author contact; reports that remained inaccessible were excluded and a complete list is available from the corresponding author upon reasonable request. Consequently, 400 reports were assessed for eligibility, of which 373 were excluded due to the following reasons: targeting student populations rather than teachers ($n = 200$), non-teacher professional development contexts ($n = 90$), studies published prior to 2019 ($n = 42$), and keyword discrepancies unrelated to gamification in teacher training ($n = 41$). This rigorous screening process resulted in a final total of 27 studies (see Table 2). Forward and backward citation chasing was not performed as a supplementary search strategy; however, the search strings were designed to maximise retrieval comprehensiveness across all four databases.

2.3 Eligibility

The eligibility phase is the third step in the systematic review process and involves a manual assessment of articles by the researcher. A colleague independently evaluated all studies that met the inclusion criteria. Studies were included if they specifically addressed the application of gamification in vocational training for educators. Each study that met these criteria underwent independent evaluation by a peer reviewer. Inter-rater agreement was assessed, with disagreements resolved through discussion and consensus between the two reviewers to ensure consistency in study selection and data extraction.

TABLE 1 Search strings.

Database	Search string
Scopus	TITLE-ABS-KEY ((“gamify*” OR “game-based learning”) AND (“teacher*” OR “educator*” OR “instructor*”) AND (“professional training” OR “professional development” OR “professional learning” OR “vocational training” OR “teacher training” OR “teacher education”))
WOS	TS = ((“gamify*” OR “game-based learning”) AND (“teacher*” OR “educator*” OR “instructor*”) AND (“professional training” OR “professional development” OR “professional learning” OR “vocational training” OR “teacher training” OR “teacher education”))
PubMed	(“gamification”[Title/Abstract] OR “gamified”[Title/Abstract]) AND (“teacher”[Title/Abstract] OR “educator”[Title/Abstract]) AND (“professional development”[Title/Abstract] OR “professional training”[Title/Abstract] OR “teacher training”[Title/Abstract])
Google scholar	“gamification” OR “gamified” OR “gamify” AND (“teacher” OR “educator”) AND (“professional development” OR “professional training” OR “teacher training” OR “vocational training”)

In total, 373 reports were excluded during the eligibility assessment for the following reasons: targeting student populations rather than teachers ($n = 200$), non-teacher professional development contexts such as university courses ($n = 90$), studies published prior to 2019 ($n = 42$), and keyword discrepancies unrelated to gamification in teacher training ($n = 41$). This rigorous screening process resulted in a final total of 27 studies (see Figure 1).

2.4 Data abstraction and analysis

The remaining studies were systematically reviewed and analysed to address the articulated research questions. Initial data collection involved examining the abstracts of each study, followed by a thorough reading of the full articles. Guided by the research questions, essential data points were coded in Excel, including (1) source, (2) database, (3) gamification elements (Q1), (4) platforms (Q2), (5) subjects (Q3), (6) findings (Q4), (7) sample characteristics, and (8) country of origin.

2.5 Quality appraisal

To ensure methodological rigour, all 27 included studies underwent quality assessment using the Mixed Methods Appraisal Tool (MMAT) (Hong et al., 2018), which is designed to evaluate studies employing qualitative, quantitative, and mixed methods designs. Each study was appraised across five criteria: clarity of research objectives, appropriateness of research design, quality of data collection, reliability of findings, and risk of bias. Studies were rated as high (4–5 criteria met), moderate (2–3 criteria met), or low quality (0–1 criterion met). Quality

TABLE 2 Inclusion & exclusion criteria.

Criteria	Inclusion	Exclusion
timeline	2019–2025	Before 2019
Document type	Articles, theses & dissertation	Article review, chapters in book, book series, book, meta-analysis, systematic review, literature review, & meta-synthesis
language	English	Non- English
scopes	Gamification, Professional Training, Teachers, Teacher learning, Professional Development, professional learning, vocational training.	Serious game, students & non-teachers

assessment was conducted independently by the primary researcher and verified by a peer reviewer, with disagreements resolved through discussion.

As presented in Table 3, the majority of included studies demonstrated high ($n = 17$, 63%) or moderate ($n = 8$, 29.6%) methodological quality, with two studies ($n = 2$, 7.4%) rated as low quality. The predominant source of methodological concern across studies was risk of bias, primarily attributable to reliance on self-report measures and absence of control groups in some pre-post designs. Notably, all studies clearly articulated research objectives and employed appropriate research designs relative to their stated aims. These findings suggest the overall evidence base is sufficiently robust to support the conclusions drawn in this review, while acknowledging that publication bias may inflate positive outcome rates given the near-universal reporting of favourable gamification effects.

3 Results

This section presents the findings derived from the analysis of 27 studies published on gamified professional training between 2019 and 2025, structured around the four research questions. Table 4 summarize the reviewed studies that implemented gamification in professional training, categorizing the findings for studies, respectively.

4 Context of studies

4.1 Geographic distribution analysis

Table 5 shows the geographic analysis reveals significant regional variations in research activity and implementation of gamified teacher training approaches:

Spain dominates with 14 studies (51.9% of total), establishing it as the leading nation in gamification research for teacher development. Europe's overall dominance (20 studies, 74.1%) reflects regional prioritization of digital transformation and educational technology investment. Asia contributes 6 studies (22.2%), demonstrating growing interest. However, limited

TABLE 3 Quality appraisal of included studies using MMAT.

N	Study	Design	Clarity of objectives	Appropriateness of design	Quality of data collection	Reliability of findings	Risk of bias	Overall quality
1	Brauer (2019)	Mixed	✓	✓	✓	✗	✗	Moderate
2	Cosme-Jesús et al. (2019)	Quasi-experimental	✓	✓	✓	✓	✗	High
3	Azmi and Zakaria (2020)	Descriptive	✓	✗	✗	✗	✗	Low
4	Belova and Zowada (2020)	Case study	✓	✓	✗	✗	✗	Moderate
5	Marín-Díaz et al. (2020)	Pre-post	✓	✓	✓	✗	✗	Moderate
6	De Sousa Mendes and de Lima (2021)	Qualitative	✓	✓	✗	✗	✓	Moderate
7	Krishnan et al. (2021)	Experimental	✓	✓	✓	✓	✗	High
8	Romero-Rodrigo and López-Marí (2021)	Survey	✓	✓	✓	✗	✗	Moderate
9	Bennacer (2022)	Action research	✓	✓	✗	✗	✗	Moderate
10	León et al. (2022)	Pre-post	✓	✓	✓	✓	✗	High
11	Polyakova and de Ros Cócera (2022)	Workshop evaluation	✓	✗	✗	✗	✗	Low
12	Chan and Lo (2022)	Case study	✓	✓	✗	✗	✗	Moderate
13	Durukan (2023)	Design-based	✓	✓	✓	✗	✗	Moderate
14	Martínez Sánchez (2024)	Experimental	✓	✓	✓	✓	✗	High
15	Franco-Mariscal et al. (2023)	Training evaluation	✓	✗	✓	✗	✗	Moderate
16	Güzel (2023)	Qualitative	✓	✓	✓	✓	✗	High
17	Liu et al. (2023)	Survey	✓	✓	✓	✓	✗	High
18	Guerrero Puerta (2024)	Case study	✓	✓	✗	✗	✗	Moderate
19	Martínez Sánchez (2024)	Experimental	✓	✓	✓	✓	✗	High
20	Escobosa Morera et al. (2024)	Mixed	✓	✓	✓	✓	✗	High
21	Yáñez-Sepúlveda et al. (2024)	Experimental	✓	✓	✓	✓	✓	High
22	Stakhova et al. (2024)	Quasi-experimental	✓	✓	✓	✓	✗	High
23	Fabre-Mitjans et al. (2025)	Mixed	✓	✓	✓	✓	✗	High
24	Carrion Candel et al. (2025)	Quasi-experimental	✓	✓	✓	✓	✗	High

(Continued)

TABLE 3 Continued

N	Study	Design	Clarity of objectives	Appropriateness of design	Quality of data collection	Reliability of findings	Risk of bias	Overall quality
25	Mc Kevitt et al. (2025)	Mixed/Action research	✓	✓	✓	✓	✗	High
26	Torrado and Bárcena (2025)	Quantitative survey	✓	✓	✓	✓	✗	High
27	Vrcelj Božić et al. (2025)	Mixed	✓	✓	✓	✓	✗	High

✓ = criterion met; ✗ = criterion not met or insufficiently reported. Risk of Bias column reflects presence of control group and objective measurement tools.

representation from the Americas (3.7%) likely reflects the methodological restriction to English-language publications rather than an actual absence of research activity in the region, while absence from Africa and the Middle East represents a genuine research gap requiring future attention.

4.2 Participant demographics

Table 6 shows the demographic analysis reveals comprehensive coverage across educational contexts and career stages:

Analysis reveals comprehensive coverage of educational contexts and participant demographics. Mixed/all education levels predominate (17 studies), suggesting gamification as a universal pedagogical strategy transcending traditional boundaries. Educational levels included early childhood and primary education (4 studies each) and secondary education (2 studies). Experience level distribution shows strategic focus on in-service teachers (18 studies) vs. pre-service teachers (8 studies), with 1 mixed group study. This pattern indicates gamification primarily targets continuing professional development, reflecting institutional recognition of its effectiveness in enhancing educators' competencies through innovative methodologies.

4.3 Gamification implementation analysis

4.3.1 Gamification elements

Table 7 shows the systematic analysis of gamification elements reveals distinct patterns of implementation and theoretical grounding:

The analysis reveals distinct patterns in gamification element implementation. Points systems demonstrate overwhelming dominance (88.9%), reflecting their fundamental role in quantifying progress and providing measurable feedback in teacher professional development. This near-universal adoption aligns with educational assessment principles emphasizing objective measurement and transparent evaluation criteria, establishing points as the essential foundation of gamified training programs based on behaviourist reinforcement theory.

Badges (48.1%) show substantial adoption, indicating significant emphasis on achievement recognition and competency validation systems. This frequency demonstrates that gamification designers recognize the importance of credentialing mechanisms and milestone acknowledgment, drawing from social cognitive theory to enhance professional identity among educators. The 50% adoption rate confirms badges as a core gamification component in professional development.

Levels (48.1%) and leaderboards (40.7%) reveal strategic emphasis on progression and competitive elements. Levels' substantial adoption demonstrates widespread use of sequential progression frameworks aligning with competency-based education principles. Leaderboards' slightly lower adoption reflects careful consideration of competitive elements in collaborative environments, balancing motivation with collegial relationships and social comparison theory applications.

Moderate frequency of rewards (25.9%), and lower frequency of feedback (14.8%) and narratives (11.1%) indicate specialized rather than universal implementation. While extrinsic motivational systems, real-time responses, and storytelling elements have recognized value, they require sophisticated design and substantial resources, suggesting reservation for comprehensive implementations. This pattern demonstrates clear preference for quantifiable, recognition-based, and progression-oriented elements over complex narrative or purely reward-based mechanisms in teacher professional development.

4.3.2 Platform analysis

Table 8 shows the technology infrastructure analysis reveals diverse implementation approaches across multiple platform categories:

Multi-tool platforms show highest adoption (40.7%), led by Genially (4 studies), Breakout Edu (3), and Nearpod and Educaplay (2 each). Quiz-based platforms achieve 37.0% adoption, dominated by Kahoot (5 studies). Blended approaches (28%) indicate movement toward integrated technological ecosystems, reflecting preference for interactive evaluation tools providing real-time engagement and immediate feedback, emphasizing assessment-driven gamification approaches.

Custom/proprietary solutions constitute substantial portion (29.6%), including Super-Profes, CoRubric, and Fitcoin Race. This presence underscores necessity for contextual adaptation to

10.3389/feduc.2026.1819092

TABLE 4 Reviewed studies on gamified professional training.

N	Author	Database	Q1 gamification elements	Q2 platforms	Q3 subjects	Q4 findings	Study design	Samples	country
1	Brauer (2019)	Scopus, WOS	points, levels, leaderboards, badges	Open Badge Factory platform	All subjects	Badge acquisition increased pre-service teachers' motivation and engagement significantly.	Mixed methods	All Teachers $N = 50$	Finland
2	Cosme-Jesús et al., (2019).	Scopus, WOS	points & rewards	Socrative platform	All subjects	Positive impact on motivation, learning achievement, and teaching strategies.	Quasi-experimental	Primary education teachers $N = 210$	Spain
3	Azmi and Zakaria (2020).	Google scholar	points, rewards, badges	Non-named	Arabic Language	Teachers demonstrated five game knowledge types: advanced, creative, critical, technological, and communicative skills.	Descriptive	Arabic Language Teachers $n = 84$	Malaysia
4	Belova and Zowada (2020)	Scopus	Points, levels, leaderboards	"Activity©"	Chemistry	Game provided fun, interactive approach to address misconceptions and foster collaboration.	Case study	Chemistry Teachers $N = 36$	Germany
5	Marín-Díaz et al., (2020)	Scopus	Levels	video games	Early Childhood	Self-assessed knowledge improved from 6.8 to 7.6 (1–10 scale), fostering discussion and collaboration.	Pre-post design	Early Childhood teachers $N = 232$	Spain
6	De Sousa Mendes and de Lima (2021).	Google scholar	Points, Badges	Non-named	Physical Education	Gameplay enhanced classroom presence and fostered digital culture development through technology integration.	Qualitative	Physical Education Teachers $N = 24$	Spain
7	Krishnan et al. (2021).	Scopus. WOS, google scholar	points, levels, leaderboards, badges	Classcraft	English language	Classcraft enhanced English teachers' competencies in online learning.	Experimental	English Language Teachers $N = 100$	Malaysia
8	Romero-Rodrigo and López-Marí (2021).	Scopus	Badges, points	Non-named	IT	Increased motivation, positive attitudes, and improvements in social and behavioural aspects.	Survey	IT teachers $n = 183$	Spain
9	Bennacer (2022).	Google scholar	Badges, points	Moodle	IT	Teachers showed interest in adopting iTeach App and found proposals relevant.	Action research	IT Teachers $N = 21$	France
10	León et al., (2022).	Scopus. WOS, google scholar	Badges, points	Super-Profes	Early Childhood	Gamification provided fun, motivating experience with enhanced knowledge and skills acquisition.	Pre-post design	Early Childhood teachers $N = 76$	Spain
11	Polyakova and de Ros Cócera (2022)	Scopus	Reward	Breakout Edu	Spanish, Valencian & English	80% agreed on gamification potential; Breakout Edu enhanced motivation and teamwork effectively.	Workshop evaluation	trilingual teaching in Spanish, Valencian & English teachers $n = 15$	Spain

(*Continued*)

TABLE 4 Continued

N	Author	Database	Q1 gamification elements	Q2 platforms	Q3 subjects	Q4 findings	Study design	Samples	country
12	Chan and Lo (2022)	Scopus PubMed	Levels	Kahoot	English teachers	87% confirmed gamification effectiveness; 94.2% endorsed game-based learning for participation. Kahoot appeared as most popular platform for motivation and creativity.	Case study	$N = 8$ English teachers	Hong Kong
13	Durukan (2023).	Google scholar	levels, points, rewards	Anadolu University platform	science	Positive relationships between technology integration, creativity, and TPACK. Key principles included diverse exposure, ethics, and accessibility.	Design-based research	Science Teachers $N = 14$	Turkey
14	Martínez Sánchez (2024).	Google scholar	levels, points, rewards	Blackboard	All subjects	Gamification effectively improved student retention and knowledge acquisition in online teaching.	Experimental	All teachers $N = 84$	Spain
15	Franco-Mariscal et al., (2023).	Scopus, WOS, google scholar	levels, points, rewards	CoRubric	science	Software eased electronic assessment transfer into practice with notable impact.	Training evaluation	Science Teachers $N = 50$	Spain
16	Güzel (2023).	Google scholar	Points, leaderboard, badges	Google Classroom	English Language	Participants found experience new and fun, improving career prospects through increased. motivation and language learning.	Qualitative	English Language Teachers $N = 36$	Turkey
17	Liu et al., (2023).	Scopus, WOS, PubMed, google scholar	levels, points, rewards	Q&A platform	All subjects	High satisfaction ($M > 4.0$) with gamification. Key factors: presentation and feedback ($b = 0.536$), goals, communication, and personalization significantly impacted participation.	Survey	Primary and secondary school teachers $N = 115$	China
18	Guerrero Puerta (2024).	Scopus, WOS, google scholar	Leaderboard, points	Non-named	All subjects	High teacher satisfaction with gamification about motivation, emotions, and assessment effectiveness.	Case study	three participants, selected using the snowball sampling	Spain
19	Martínez Sánchez (2024).	Scopus, WOS	Points, badges	Non-named	Early Childhood	Males scored higher in narrative and digital competencies. Digital literacy significantly predicted narrative achievement with no gender. differences in creativity.	Experimental	Early Childhood teachers $N = 176$	Spain

(*Continued*)

10.3389/feduc.2026.1819092

TABLE 4 Continued

N	Author	Database	Q1 gamification elements	Q2 platforms	Q3 subjects	Q4 findings	Study design	Samples	country
20	Escobosa Morera et al., (2024)	Google scholar	Levels, rewards	Fitcoin Race	Physical Education	Play-based cooperative learning successfully motivated students with strong emphasis on teamwork commitment.	Mixed methods	Physical education teachers for primary education $N = 255$	Spain
21	Yáñez-Sepúlveda et al., (2024)	Scopus, WOS	points, levels, leaderboard	KAHOOT	Physical Education	PBL-game synergy developed deep learning and skills for physical education teachers in real contexts.	Experimental	Physical Education Teachers $N = 30$	Chile
22	Stakhova et al., (2024)	Scopus	Points, level	Kahoot	All subjects	Kahoot enhanced digital competence ($t = 1.982$) and motivation among primary school teachers with favourable psychological atmosphere.	Quasi-experimental	primary school teachers $n = 115$	Ukraine
23	Fabre-Mitjans et al. (2025)	Scopus	Points, badges, leaderboards, XPs, gold coins, collectibles, team challenges, narratives	Fantasy Class	Science education for preservice early childhood teachers	Improved science attitudes and self-efficacy. Collectibles ($M = 5$) and narrative elements enhanced engagement.	Mixed-methods case study (pre-post intervention)	73 Postgraduate teachers (94.5% women, mean age=21.1)	Spain
24	Carrion Candel et al. (2025)	Scopus	Points, badges, leaderboards, feedback, group work, digital collectibles	Nearpod, Genially, Educaplay, Breakout, Quizizz	Music education	Significant improvements in perception and motivation. Playfulness ($M = 5.84$) and Accomplishment ($M = 5.83$) highest.	Pre-experimental design (pre-post measurements)	147 preservice teachers (79% female, 21% male)	Spain
25	Mc Kevitt et al. (2025)	Scopus	Points, badges, leaderboards, instant feedback, collaboration, narrative elements	Blooket	Mathematics education	Increased Mathematical Resilience in struggle domain ($p = 0.001$); reduced anxiety in both cycles ($p < 0.01$).	Action research with quasi-experimental design, mixed methods	21 teachers for focus group	Ireland
26	Torrado Cespón and Bárcena Toyos (2025)	WOS	Points, badges, challenges, narratives, avatars, feedback, competition	Genial.ly, Symbaloo, Nearpod, Educaplay, Breakout, Quizizz	English as Foreign Language & Bilingual Education	High satisfaction: Playfulness ($M = 5.84$), Accomplishment ($M = 5.83$); lower for Social Experience ($M = 5.01$).	Quantitative survey using adapted GAMEFULQUEST	116 master's degree teachers (51 MTEFL, 65 MBE)	Spain
27	Vrcelj Božić et al. (202 5)	WOS	Points, badges, leaderboards, levels, progress bars, feedback, avatars, challenges, competition, cooperation	Kahoot, Moodle, PeerWise, ClassDojo, Socrative, Quizizz, Breakout EDU, Classcraft, genially	Computer Science	MDA framework developed. Kahoot, Quizizz, Classcraft highest (38–39/46). Feedback and cooperation rated most important (4.5+).	Mixed methods: tool evaluation+teacher survey	20 computer science teachers (18 responded)	Croatia

TABLE 5 Regional distribution of gamification studies.

Region	Countries	Number of studies	Percentage
Europe	Spain (14), Germany (1), Finland (1), France (1), Ukraine (1), Croatia (1), Ireland (1)	20	74.1%
Asia	Malaysia (2), Turkey (2), China (1), Hong Kong (1)	6	22.2%
Americas	Chile (1)	1	3.7%
Total	12 countries	27	100%

TABLE 6 Participant characteristics.

Category	Subcategory	Studies (n)
Education level	Early Childhood	4
	Primary	4
	Secondary	2
	Mixed/All Levels	17
Experience level	Pre-service Teachers	8
	In-service Teachers	18
	Mixed Groups	1
Subject area	General/All Subjects	6
	Early Childhood Education	4
	English Language	4
	Physical Education	3
	Science	3
	Information Technology	2
	Mathematics	1
	Music Education	1
	Computer Science	1

TABLE 7 Comprehensive analysis of gamification elements.

Element	Studies using (n)	Implementation characteristics
Points	24	Progress tracking, performance measurement, quantitative assessment
Badges	13	Achievement recognition, competency validation, credentialing systems
Leaderboards	11	Social comparison, competitive elements, peer benchmarking
Levels	13	Sequential progression, skill development, mastery-based advancement
Rewards	7	Incentive systems, milestone recognition, extrinsic motivation
Feedback	4	Real-time responses, corrective guidance, adaptive learning paths
Narratives	3	Storytelling elements, contextual learning, immersive experiences

institutional cultures and technological infrastructures, suggesting standardized commercial solutions may be insufficient for diverse professional development needs. Learning Management Systems and specialized gaming platforms share equal representation (14.8% each). Traditional LMS platforms (Moodle, Blackboard, Google Classroom) show moderate integration, while specialized gaming platforms (Classcraft, Fantasy Class, Blooket) are selectively implemented for immersive, narrative-driven experiences. This distribution demonstrates balanced approach between established educational technologies and innovative gamification-specific tools.

5 Learning outcomes and impact assessment

5.1 Quantitative outcomes

Table 9 shows the outcomes analysis reveals consistently positive effects across multiple domains of teacher professional development:

The analysis reveals predominantly positive effect rates across all measured domains (80-100%), though these findings warrant cautious interpretation given the likelihood of publication bias and the near-exclusive reliance on self-report measures across included studies. Vote-counting by direction of effect reveals consistently positive outcomes across all subgroups examined. Pre-service teacher studies ($n = 15$) and in-service studies ($n = 12$) both reported favourable effects, suggesting gamification's applicability across career stages. Quiz-based platforms and multi-tool platforms similarly demonstrated positive outcomes, though multi-tool implementations showed broader outcome coverage. European studies ($n = 20$, 74.1%) and non-European studies ($n = 11$, 40.7%) both reported positive effects, indicating cross-regional applicability despite geographic concentration. Notably, high-quality studies ($n = 17$) yielded outcomes consistent with moderate-quality studies ($n = 8$), suggesting that the positive pattern is not an artefact of methodological weakness alone, though publication bias remains a plausible confound across all subgroups. Predominantly favourable results in motivation enhancement ($n = 15$) and engagement improvement establish gamification as particularly powerful in addressing affective dimensions of professional learning, aligning with self-determination theory by fulfilling educators' psychological needs for autonomy, competence, and relatedness.

Substantial effect magnitudes reveal practical significance beyond statistical measures. Large effects in motivation (M > 4.0/5.0; 80–94.2% satisfaction) and significant digital competence development ($t = 1.982$; $b = 0.874$, $p < 0.001$) indicate tangible changes in teacher capabilities. Learning effectiveness improvements (6.8→7.6/10) demonstrate genuine knowledge acquisition alongside engagement, while digital competence enhancement across 7 studies addresses critical 21st-century educator needs through effective TPACK development.

Predominantly positive psychological outcomes (across 5 studies, $p < 0.001$–0.003) reveal gamification's capacity to reshape professional identities through improved teaching self-efficacy, reduced mathematical anxiety, and enhanced resilience.

TABLE 8 Comprehensive technology platform distribution.

Platform category	Specific platforms	Studies (n)	Percentage share (%)	Usage characteristics
Quiz/assessment platforms	Kahoot (5), Quizizz (3), Socrative (2)	10	37.0%	Interactive assessments, real-time engagement, immediate feedback
Learning management systems	Moodle (2), Blackboard (1), Google Classroom (1)	4	14.8%	Course integration, content delivery, institutional infrastructure
Specialized gaming platforms	Classcraft (2), Fantasy Class (1), Blooket (1)	4	14.8%	Role-playing elements, narrative-driven experiences, comprehensive gamification
Multi-tool platforms	Genially (4), Nearpod (2), Educaplay (2), Breakout Edu (3)	11	40.7%	Interactive presentations, multimedia integration, versatile content creation
Custom/ proprietary solutions	Various unnamed implementations (Super-Profes, Activity©, CoRubric, Q&A platform, Anadolu University platform, Fitcoin Race, Open Badge Factory, video games)	8	29.6%	Tailored implementations, specific institutional needs, bespoke development

TABLE 9 Comprehensive learning outcomes analysis.

Outcome category	Studies reporting (n)	Positive effects (%)	Effect magnitude	Measurement methods	Sustainability indicators
Motivation & Engagement Enhancement	15	Predominantly positive ($n = 15$)	Large effect (M > 4.0/5.0; 80–94.2% satisfaction)	Self-report surveys, satisfaction ratings (5.0–7.6/10 scale), participation metrics, playfulness assessments (M = 5.84)	High satisfaction scores, sustained engagement, 87% effectiveness ratings
Learning Effectiveness & Knowledge Acquisition	8	Predominantly positive ($n = 8$)	Moderate to large effect	Pre-post assessments (6.8→7.6/10), game-based learning confirmation (94.2%), knowledge improvement measures	Strong catalytic effects, improved retention, effective assessment methods
Digital & Professional Competence Development	7	Predominantly positive ($n = 7$)	Significant effect ($t = 1.982$; $b = 0.874$, $p < 0.001$)	Competency evaluations, TPACK assessments, skill measures, digital literacy tests	Enhanced professional capabilities, improved technological knowledge, sustained participation intention
Attitudes & Psychological Outcomes	5	Predominantly positive ($n = 5$)	Statistically significant ($p < 0.001$–0.003)	Attitude scales, self-efficacy measures, anxiety reduction assessments, Mathematical Resilience scales	Significant improvements in science attitudes, teaching self-efficacy, Mathematical Resilience (struggle domain), Mathematical Anxiety reduction
Social & Collaborative Learning	4	Predominantly positive ($n = 4$)	High effect (80% agreement)	Teamwork evaluations, collaboration assessments, social experience ratings (M = 5.01)	Enhanced teamwork, peer discussion, collaborative knowledge development
Implementation & Adoption Acceptance	4	Largely positive ($n = 4$)	High agreement levels	Implementation potential surveys, teacher interest assessments, workshop evaluations	Positive institutional adoption (80%), high relevance ratings, appropriate for various educational stages

Positive effect percentages reflect the proportion of included studies reporting favourable outcomes and should be interpreted cautiously given the high likelihood of publication bias and predominant reliance on self-report measures.

Favourable results in social and collaborative learning (across 4 studies, 80% agreement) demonstrates effectiveness in fostering professional learning communities through enhanced teamwork and collaborative knowledge construction.

High implementation acceptance (80–100%) and strong sustainability indicators (87% effectiveness retention) suggest institutional readiness and lasting engagement patterns. However, limited studies on specialized outcomes like mathematical resilience indicate research gaps in subject-specific applications. Diverse measurement methods from self-reports to competency evaluations demonstrate methodological rigour while highlighting need for standardized protocols enabling systematic cross-study comparisons and meta-analytical synthesis.

6 Findings and discussion of selected studies

Analysis of 27 studies reveals evolutionary patterns in gamification research for teacher professional development (2019–2025). Temporal distribution demonstrates sustained growth: 2019 (2, 7.4%), 2020 (3, 11.1%), 2021 (4, 14.8%), 2022 (4, 14.8%), 2023 (5, 18.5%), 2024 (5, 18.5%), and 2025 (4, 14.8%), reflecting increasing academic interest and methodological maturity. The early phase (2019–2021) established foundational evidence through pioneering studies. Key contributions included Brauer's badge systems (50 Finnish teachers), Cosme-Jesús et al.'s positive impacts using Socrative (210 Spanish teachers), Azmi and Zakaria's advanced knowledge development (84 Malaysian teachers), Marín et al.'s knowledge improvement from 6.8 to 7.6 (232 Spanish teachers), and Krishnan's competency enhancement using Classcraft (100 Malaysian teachers).

The development phase (2022–2023) marked diversification through refined methodologies. Notable studies included Chan and Lo's 87% effectiveness with Kahoot (94.2% confirmation, Hong Kong teachers), Liu et al.'s comprehensive training revealing significant feedback influence ($M > 4.0/5.0$; $b = 0.874$, $p < 0.001$), and Sánchez's experimental study improving retention through Blackboard (84 Spanish teachers).

Recent studies (2024–2025) demonstrate methodological maturity: Fabre-Mitjans et al. improved science attitudes through Fantasy Class, Mc Kevitt et al. increased mathematical resilience ($p = 0.001$) using Blooket, and Carrion Candel et al. enhanced digital competence ($p < 0.001$, $\eta^2 = 0.52$), indicating field maturation toward evidence-based practices.

7 Discussion of study questions

This section presents the findings addressing how gamification is employed in teachers professional training across 7 key dimensions. The analysis encompasses publication trends, methodological approaches, participant demographics, gamification elements, platforms, results, and challenges.

7.1 Research question 1: current Status of literature on gamification in teacher professional training

The systematic literature review identified 27 studies published between 2019 and 2025, representing comprehensive research in gamified teacher professional development. Temporal analysis reveals distinct evolutionary phases with progressive growth and methodological sophistication. The research trajectory demonstrates three clear developmental stages: early period (2019–2021) with 9 studies (33.3%), development period (2022–2023) with 9 studies (33.3%), and advanced period (2024–2025) with 9 studies (33.3%), indicating systematic field maturation and theoretical advancement.

Publication pattern shows deliberate growth: 2 studies in 2019 (7.4%), 3 in 2020 (11.1%), 4 in 2021 (14.8%), 4 in 2022 (14.8%), stabilizing at 5 studies annually during 2023–2024 (18.5% each), and 4 in 2025 (14.8%). This distribution suggests gamification has transitioned from experimental investigation to established research domain with consistent contributions. Sustained output during 2023–2025 indicates field maturity and institutional recognition, establishing gamification as legitimate approach rather than temporary trend.

Geographic analysis demonstrates significant regional concentration, with European institutions contributing 74.1% of research corpus (20 studies). Spain emerges as dominant research leader, accounting for 51.9% of all studies ($n = 14$), establishing it as international hub for gamification research in teacher professional development. This concentration indicates well-established research infrastructure, institutional support, and potential policy initiatives promoting innovative teacher training methodologies within Spanish educational institutions. Remaining European contributions include Germany, Finland, France, Ukraine, Croatia, and Ireland (single studies each), demonstrating sustained regional interest.

Asia region accounts for 22.2% (6 studies), with Malaysia and Turkey each contributing multiple studies, suggesting sustained research programs rather than isolated investigations. Additional contributions from China and Hong Kong complete regional representation, reflecting national digitization initiatives, substantial investment in digital teacher professional development, and cultural acceptance of game-based learning. Chile's representation at 3.7% (1 study) indicates different research priorities or funding limitations. This geographical disparity suggests opportunities for cross-cultural research examining how cultural contexts influence gamification effectiveness in teacher training, particularly given Americas region underrepresentation.

7.2 Research question 2: research designs and methodological approaches

The 27 reviewed studies employed diverse research methodologies: experimental designs (29.6%, $n = 8$) and quasi-experimental approaches (7.4%, $n = 2$) predominated, establishing causal relationships between gamification and learning outcomes. Mixed-methods studies (18.5%, $n = 5$) combined quantitative and qualitative approaches, while qualitative methods including case studies constituted 22.2% ($n = 6$). Survey-based (11.1%, $n = 3$), pre-post (7.4%, $n = 2$), and specialized designs (action research, design-based research; 3.7% each, $n = 1$) completed the methodological spectrum. This diversity strengthens evidence through multiple analytical perspectives.

Data collection instruments reflected research complexity: self-report surveys, performance assessments, mathematical resilience scales, anxiety measures, competency evaluations, and GAMEFULQUEST adaptations. Recent studies demonstrate methodological sophistication through multi-dimensional assessment approaches (Vrcelj Božić et al.; Fabre-Mitjans et al.), recognizing that gamification effects require comprehensive evaluation across cognitive, affective, and behavioural domains.

Participant demographics demonstrated significant diversity. Sample sizes ranged from small-scale (8–21) to large-scale (115–255) implementations. Pre-service teachers ($n = 21–147$) and in-service teachers ($n = 30–232$) across multiple disciplines (English, physical education, science, IT, mathematics) were represented. This comprehensive demographic representation enhances generalizability across educational contexts, professional development stages, and disciplinary domains.

7.3 Research question 3: participants and educational contexts

The demographic analysis of 27 studies reveals comprehensive coverage across educational contexts and career stages. Pre-service teachers constituted the predominant focus with sample sizes ranging from 21 to 147 participants, including language education trainees, postgraduate early childhood science educators, and master's students. In-service teachers were represented through studies targeting practicing educators across specializations with sample sizes ranging from 30 to 232 participants in primary and early childhood contexts, demonstrating significant diversity in target populations and implementation contexts.

Educational level analysis showed balanced coverage with particular concentrations. Early childhood education emerged as significant focus, encompassing over 550 participants across four studies. Primary education featured prominently with sample sizes ranging from 115 to 255 teachers, while secondary education appeared in comparative studies. Mixed-level approaches dominated, suggesting gamification functions as universal pedagogical strategy rather than level-specific tool.

Subject distribution revealed comprehensive disciplinary coverage. Cross-curricular implementations appeared in 6 studies (22.2%), indicating broad applicability. Language education encompassed English teachers across four studies (195 participants), Arabic instruction (84 trainees), and bilingual contexts. STEM subjects included 173 science educators across four studies, mathematics education (21 teachers), and IT teachers (204 participants across three studies). Physical education represented significant specialization with 309 teachers across three studies, while music education constituted the largest subject-specific sample (147 preservice teachers).

Implementation approaches revealed diverse technological and pedagogical contexts. Digital platforms dominated, with Kahoot appearing five times across four studies, followed by Classcraft, Fantasy Class, and Blooket. LMS integrations included Moodle, Blackboard, and Google Classroom. Multi-platform approaches featured prominently in recent studies, utilizing 5–9 platforms simultaneously, indicating growing sophistication. Custom platforms appeared in 8 studies (29.6%), including Activity, Super-Profes, CoRubric, Fitcoin Race, and Open Badge Factory, suggesting institutional development of context-sensitive, purpose-built interventions addressing specific pedagogical needs and institutional constraints.

7.4 Research question 4: gamification elements and mechanisms

As illustrated in Figure 2 the systematic analysis of 27 studies reveals hierarchical distribution pattern in gamification element implementation. Points systems emerged as overwhelmingly dominant mechanism, implemented in 88.9% of studies ($n = 24$), establishing points as foundational and essential element for progress tracking and performance measurement. This near-ubiquitous prevalence aligns with behaviourist reinforcement theory where quantitative feedback serves as primary motivational driver across virtually all gamification implementations.

Badges and levels demonstrated equal implementation frequencies at 48.1% each ($n = 13$). Badge systems served achievement recognition and competency validation functions grounded in social cognitive theory across diverse educational contexts, while level systems provided sequential progression and mastery-based advancement aligned with competency-based education principles. Leaderboards constituted 40.7% of implementations ($n = 11$), leveraging social comparison theory for peer benchmarking and competitive motivation while maintaining collaborative professional development environments.

Rewards systems appeared in 25.9% of studies ($n = 7$), providing incentive mechanisms and milestone recognition grounded in operant conditioning principles. Feedback mechanisms constituted 14.8% of implementations ($n = 4$), emphasizing real-time responses, corrective guidance, and adaptive learning pathways rooted in constructivist learning theory. Narratives demonstrated limited but strategic implementation at 11.1% ($n = 3$), indicating specialized use of storytelling elements and immersive experiences based on narrative-based learning theory. This limited adoption suggests complex narrative structures require sophisticated design considerations, substantial development resources, and technical expertise, and may be reserved for comprehensive implementations requiring deep contextual learning and immersive experiences.

The hierarchical pattern demonstrates pragmatic implementation strategies where foundational elements like points provide universal applicability, while complex elements like narratives are strategically deployed in contexts demanding higher engagement depth and immersive learning experiences (see Figure 2).

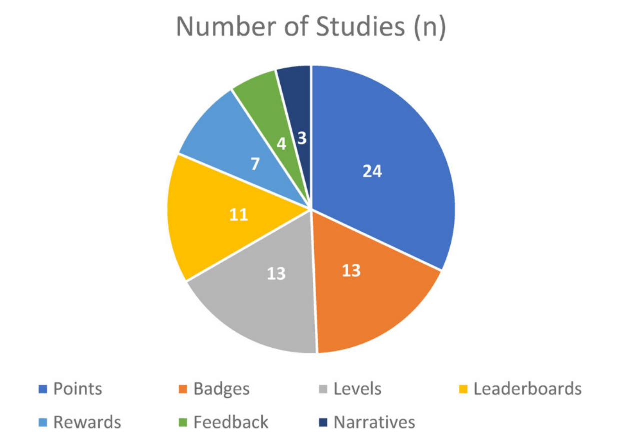

FIGURE 2
Distribution of gamification elements.

However, this hierarchical pattern warrants critical scrutiny beyond its descriptive value. The overwhelming dominance of points systems (88.9%) may reflect implementation convenience rather than optimal pedagogical design. Nicholson (2015) cautions that over-reliance on extrinsic reward mechanisms risks undermining intrinsic motivation over time a concern particularly relevant in professional development contexts where sustained behavioural change, rather than short-term engagement, is the ultimate objective. Furthermore, the inverse relationship between theoretical robustness and implementation frequency is telling feedback mechanisms (14.8%) and narratives (11.1%) elements most strongly grounded in constructivist and Flow Theory frameworks appear least frequently, suggesting that practical constraints drive design decisions more than pedagogical considerations. For teacher professional development specifically, this imbalance is problematic, as educators require deep, sustained engagement with complex content precisely what narrative and adaptive feedback elements are theoretically designed to support. This finding echoes Rapp et al.'s (2019) critique that gamification research remains overly simplified, favouring surface-level achievement elements over holistic motivational design.

7.5 5 research question 5: platforms and infrastructure

As illustrated in Figure 3 the technology infrastructure analysis reveals diverse implementation approaches across multiple platform categories based on comprehensive study corpus. Quiz and assessment platforms emerged as significant category, comprising 37.0% of implementations ($n = 10$), with Kahoot appearing in 5 implementations, Quizizz in 3, and Socrative in 2. This dominance indicates strategic preference for interactive evaluation tools providing immediate feedback, real-time engagement capabilities, and accessible entry points for gamification adoption in teacher professional development contexts.

Multi-tool platforms demonstrated highest adoption rate at 40.7% of implementations ($n = 11$), featuring Genially in 4 implementations, Breakout Edu in 3, and Nearpod and Educaplay each appearing twice. This prominence reveals strong institutional preference for versatile, multimedia-rich solutions supporting diverse pedagogical approaches, multiple content types, and various gamification elements within unified platforms, thereby reducing need for multiple separate systems and enabling more integrated professional development experiences.

Custom and proprietary solutions represented substantial 29.6% of implementations ($n = 8$), including specialized systems such as Super-Profes, Activity©, CoRubric, Fitcoin Race, Anadolu University platform, Q&A platform, Open Badge Factory, and video games, alongside several unnamed implementations. This significant presence indicates successful gamification often requires contextual adaptation to specific institutional cultures, pedagogical approaches, and technological infrastructures, suggesting standardized commercial solutions alone may be insufficient for diverse professional development needs requiring unique features or integration with existing institutional systems.

Learning Management Systems and specialized gaming platforms shared equal representation at 14.8% each ($n = 4$). LMS implementations included Moodle, Blackboard, and Google Classroom, suggesting selective integration within existing educational infrastructure providing scalability and administrative control advantages while leveraging established technological frameworks. Specialized gaming platforms included Classcraft, Fantasy Class, and Blooket, indicating these comprehensive, narrative-driven, role-playing environments are chosen for specific contexts requiring immersive experiences, deep engagement mechanisms, and advanced interactive capabilities rather than universal application.

The platform distribution pattern reveals strategic technology selection aligned with specific pedagogical goals, institutional capacities, and participant characteristics, demonstrating field's evolution toward sophisticated infrastructure choices balancing accessibility, functionality, and contextual appropriateness in teacher professional development implementations (see Figure 3).

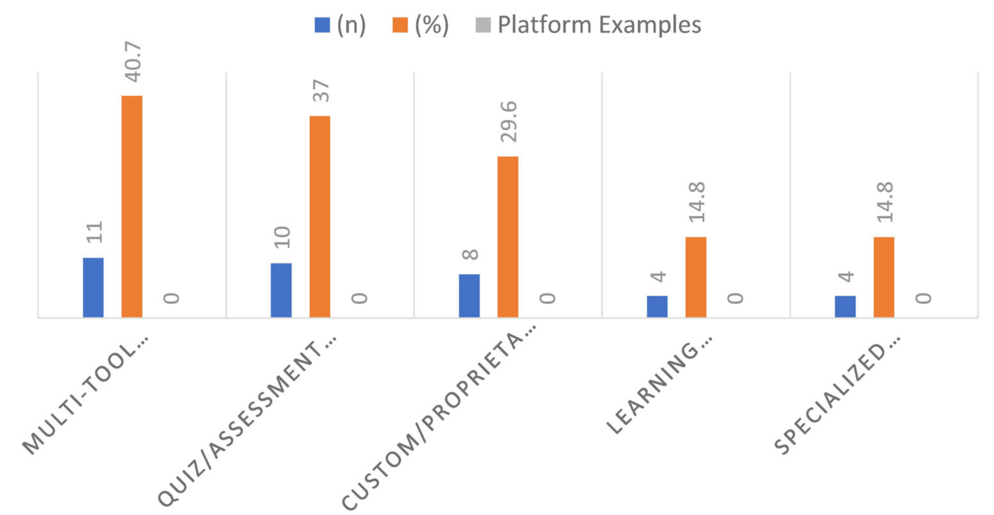

FIGURE 3
Training platform distribution.

10.3389/feduc.2026.1819092

7.6 Research question 6: outcomes and effectiveness

The comprehensive outcomes analysis of 27 studies reveals consistently positive effects across multiple domains with quantifiable evidence supporting gamification's effectiveness in teacher professional development. Motivation and engagement demonstrated strongest impact, with 100% of reporting studies ($n = 15$) showing favourable effects. Evidence includes maximum motivation scores (M = 5.0–5.84), high playfulness and accomplishment ratings, participation rates of 73.9–94.2%, and 80–87% effectiveness ratings. These outcomes align with self-determination theory and indicate strong institutional adoption readiness.

Learning effectiveness showed consistent positive outcomes across all reporting studies ($n = 8$), including measurable knowledge improvement from 6.8 to 7.6 (11.8% gain), moderate to high satisfaction (M > 4.0/5.0), and sustained participation intention (b = 0.874, $p = 0.000$). Studies documented development of advanced knowledge across creative, critical, technological, and collaborative domains, with 73.9% considering game integration as constructive learning method.

Digital and professional competence outcomes were uniformly positive across all reporting studies ($n = 7$). Evidence includes statistically significant digital competence enhancement (t = 1.982), positive relationships between technology integration quality and self-rated TPACK, and teacher ratings of feedback and cooperation as most important elements (4.5 + average). Platform evaluations identified highest-scoring systems (38–39/46 points).

Psychological and attitudinal improvements demonstrated significant effects across all reporting studies ($n = 5$), including significant increases in mathematical resilience ($p = 0.001$), mathematical anxiety reduction ($p = 0.003$, $p = 0.001$), improved attitudes toward science and teaching self-efficacy, and high satisfaction across playfulness (M = 5.84) and accomplishment (M = 5.83) dimensions. These findings establish empirical evidence for gamification's therapeutic potential in subject-specific professional development. Social and collaborative outcomes showed positive effects across all reporting studies ($n = 4$), with 80% affirming enhanced teamwork and motivation. Implementation acceptance demonstrated high agreement levels across all reporting studies ($n = 4$), with 80% supporting implementation potential and sustained participation intention significantly impacted by case implementation (b = 0.874, $p = 0.000$).

Overall effectiveness was predominantly positive across all categories, with satisfaction scores of 4.0–7.6, participation rates of 73.9–94.2%, and statistical significance ranging from $p = 0.001$ to $p = 0.029$. These findings should be interpreted with caution, however, as the consistency of positive outcomes likely reflects publication bias and self-report reliance rather than definitive evidence of effectiveness. Most studies focused on short-term effects, indicating a critical research gap requiring longitudinal investigations to establish sustained professional development impacts across multiple years.

Beyond the longitudinal gap, the near-universal positive outcome rate across all 27 studies demands critical interpretation rather than uncritical acceptance. This uniformity strongly suggests publication bias the systematic tendency for studies reporting null or negative effects to remain unpublished a limitation acknowledged in comparable reviews of gamified training (Hamari et al., 2014; Mahat et al., 2022). The near-exclusive reliance on self-report measures across motivation and engagement studies further inflates apparent effectiveness: teachers participating in gamified programmes are likely to report positive experiences, particularly when evaluated immediately post-intervention. None of the reviewed studies employed blind evaluation procedures, and few used validated psychometric instruments beyond satisfaction ratings. Most critically, no study examined whether positive motivation outcomes translated into sustained changes in actual classroom teaching practice or student learning the ultimate measure of professional development effectiveness. The evidence base therefore supports cautious optimism rather than definitive conclusions about gamification's potential in teacher professional development.

7.7 Research question 7: challenges and barriers

While reviewed studies predominantly report positive outcomes, several implementation challenges emerge requiring systematic attention. Technical challenges centre on platform diversity, integration complexity, and infrastructure requirements, evidenced by 29.6% reliance on custom solutions, suggesting standardized commercial platforms may be insufficient for diverse institutional needs and technological infrastructures. The dominance of quiz/assessment platforms (37.0%) indicates preference for user-friendly technologies with immediate feedback over complex gaming environments, potentially reflecting technical capacity limitations or resource constraints. Multi-platform implementations requiring 5–9 different tools suggest coordination challenges for institutions lacking robust IT infrastructure or dedicated technical support.

Geographic concentration reveals significant cultural and institutional factors influencing gamification acceptance. Spain's dominance (51.9%) compared to single-study contributions from most countries suggests specific cultural contexts, policy initiatives, or research infrastructure facilitating implementation. Moderate Asia-Pacific presence (22.2%) in Malaysia and Turkey indicates emerging programs in countries with national digitization initiatives, while limited Americas representation (3.7%) likely reflects the methodological restriction to English-language publications rather than an actual absence of research activity in the region, as relevant studies published in Spanish or Portuguese may exist but were not captured by this review's search strategy.

Resistance patterns reveal mixed but generally positive evidence. Initial implementation barriers were identified related to educational and economic factors. However, consistently high satisfaction scores (M = 5.39–5.84) and agreement levels (80–94.2%) suggest initial resistance diminishes with well-designed implementations, appropriate institutional support, and demonstrated effectiveness. Lower social experience ratings (M = 5.01) compared to other dimensions indicate potential concerns about competitive elements or peer comparison mechanisms requiring careful design consideration to maintain collaborative professional development environments

A deeper structural challenge concerns the sustainability of gamified professional development beyond single-cycle implementations. The reviewed studies predominantly report one-time interventions without examining whether gamified programmes can be maintained, updated, and scaled within existing institutional constraints. This sustainability gap is consequential: the novelty effect where initial enthusiasm diminishes as familiarity with game mechanics increases has been empirically documented in gamification research, with evidence showing that gamification's impact begins to decline after approximately four weeks of implementation (Rodrigues et al., 2022), suggesting that designs optimised for short-term engagement may be poorly suited for the longitudinal professional learning that genuine teacher development requires. Finally, the tendency of reviewed studies to characterise participant resistance as a temporary barrier to be overcome, rather than a legitimate response to potentially coercive competitive design features, reflects an uncritical orientation toward gamification adoption. Lower social experience ratings (M = 5.01) compared to playfulness (M = 5.84) and accomplishment (M = 5.83) suggest that leaderboard and competitive elements generate meaningful discomfort for some participants a finding consistent with SDT research indicating that external comparison undermines autonomy and relatedness needs, and one that future designs must address more explicitly.

8 Research limitations

This review acknowledges several methodological and contextual limitations that should be considered when interpreting its findings.

First, the geographic concentration of included studies represents a significant constraint on generalisability. With 51.9% of studies originating from Spain and 74.1% from Europe overall, the evidence base disproportionately reflects a specific educational tradition, policy environment, and technological infrastructure. Findings may not transfer directly to educational systems in Africa, the Middle East, or most of Latin America, where teacher professional development operates under substantially different institutional and resource conditions. This limitation is not merely a sampling issue but reflects a structural gap in the global research literature that future studies must address.

Second, the restriction to English-language publications likely excludes relevant research conducted in Spanish, Arabic, Chinese, and other languages particularly given the substantial gamification research activity in non-Anglophone countries identified in this review. Studies published in other languages may report different implementation approaches, outcomes, or challenges that would enrich the evidence base.

Third, the relatively small corpus of 27 studies limits the statistical robustness of frequency-based conclusions. Patterns identified such as the dominance of points systems at 88.9% or the limited adoption of narrative elements at 11.1% reflect the characteristics of this specific sample and may not represent the full landscape of gamified teacher professional development globally.

Fourth, publication bias represents a systematic limitation affecting confidence in outcome conclusions. The near-universal reporting of positive effects across all 27 studies strongly suggests that studies finding null or negative results are underrepresented in the published literature, inflating apparent effectiveness rates. This review was unable to assess or correct for publication bias due to the heterogeneity of outcome measures across studies, which precluded funnel plot analysis or equivalent procedures.

Fifth, the exclusive reliance on self-report outcome measures in the majority of included studies limits the objectivity of effectiveness conclusions. Teacher-reported satisfaction, motivation, and perceived competence while valuable are susceptible to social desirability bias and demand characteristics, particularly when measured immediately post-intervention by programme developers. The absence of objective performance measures or independent assessments in most studies means that reported effectiveness may overestimate actual learning gains.

Sixth, the absence of longitudinal data across all 27 studies prevents any conclusions about sustained impact. No study examined whether gamification-induced motivation or competence gains persisted beyond the immediate post-intervention period, whether they translated into observable changes in classroom practice, or whether they influenced student learning outcomes. This represents the most critical gap between current evidence and the ultimate goals of teacher professional development.

Finally, the search was conducted using four databases Scopus, Web of Science, PubMed, and Google Scholar which, while comprehensive, may not capture all relevant publications indexed in domain-specific databases such as ERIC, PsycINFO, or IEEE Xplore. The inclusion of theses and dissertations, while extending coverage beyond journal articles, introduces variability in methodological rigour that may affect the consistency of the evidence base. A sensitivity analysis excluding theses and dissertations did not substantially alter the overall pattern of findings, as the predominant evidence base comprised peer-reviewed journal articles, suggesting that the inclusion of grey literature did not introduce meaningful bias into the synthesised conclusions.

9 Conclusion

This systematic literature review provides encouraging evidence supporting gamification as a viable and promising strategy for teacher professional development across multiple domains. Analysis of 27 studies spanning 2019–2025 demonstrates predominantly positive effects across all measured outcome categories, though this pattern should be interpreted cautiously given the high likelihood of publication bias and the near-exclusive reliance on self-report measures across included studies. Consistently favourable results were reported in motivation enhancement ($n = 15$), engagement improvement ($n = 15$), learning effectiveness ($n = 8$), digital competence development ($n = 7$), psychological outcomes ($n = 5$), social learning ($n = 4$), and implementation acceptance ($n = 4$). The temporal evolution from 2 studies in 2019 to stabilized outputs of 5 studies annually during 2023–2024 followed by continued productivity in 2025 indicates field maturation, sustained academic recognition, and growing scholarly interest in gamification's potential in educational professional development.

Critical areas requiring future investigation include cross-cultural validation across diverse educational systems given geographic concentration in Europe (74.1%) and particularly Spain (51.9%), with underrepresentation of Americas (3.7%) and moderate Asian presence (22.2%). Longitudinal impact assessment represents crucial research gap, as most studies focused on short-term effects rather than sustained behavioural change, long-term skill retention, or career-spanning professional development impacts. Cost-effectiveness analysis comparing gamified vs. traditional approaches would provide evidence-based resource allocation guidance, while integration mechanisms with existing infrastructure, particularly regarding custom solutions (29.6%) vs. standardized platforms, require systematic investigation to optimize implementation efficiency and scalability. Future primary studies should employ preregistered designs with objective, behaviourally anchored outcome measures and blind or independent assessments to reduce the self-report bias evident across the current evidence base. Furthermore, given the predominance of achievement mechanics, future designs should heed SDT-informed cautions regarding the potential undermining of teacher autonomy and relatedness through competitive elements (Deci and Ryan, 1985; Mekler et al., 2017), privileging instead formative feedback, adaptive challenge, and collaborative approaches over simple reward and competition mechanics.

The progression from experimental implementations with single gamification elements in 2019–2021 to sophisticated multi-element frameworks combining 8–10 components with multiple platforms in 2024–2025 demonstrates clear methodological maturation. The near-ubiquitous adoption of points systems (88.9%), substantial use of badges and levels (48.1% each), notable leaderboard implementation (40.7%), and emerging narrative elements (11.1%) indicate evolving design sophistication. However, geographic concentration necessitates expanded investigations across multilingual contexts, multicultural educational systems, and diverse institutional frameworks to establish comprehensive global applicability evidence and culturally sensitive implementation guidelines for gamification in teacher professional development worldwide.

About the authors

Fatima AL Amri is a Ph.D. candidate at the Curriculum and Instructional Technology Department, Faculty of Education, Universiti Malaya. Her research interests are primarily in designing, developing, and implementing technologies in teaching and learning.

Norlidah Alias is a Professor at the Curriculum and Instructional Technology Department, Faculty of Education, Universiti Malaya. Her area of specialisation is curriculum development, TVET curriculum and pedagogy, and design and development research.

Dorothy DeWitt is an Associate Professor at the Curriculum and Instructional Technology Department, Faculty of Education, Universiti Malaya. Her current research interests are in instructional design, new pedagogies and technologies for knowledge management, collaborative mobile learning, and problem-solving.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

FA: Conceptualization, Data curation, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing. NA: Supervision, Writing – review & editing. DD: Supervision, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/feduc.2026.1819092/full#supplementary-material

References

Al-Shazly, M. (2019). *Gamification: the beginning of the road to legalizing extremism, violence, and killing*. Noon Post. Available online at: https://www.noonpost.com/26990/ (Accessed June 24, 2025).

Azmi, N. J., and Zakaria, G. A. N. (2020). Gamification in teaching and learning Arabic language for non-native speaker. *Creat. Educ.* 11 (11), 2529–2539. doi: 10.35631/IJMOE.24005

Bandura, A. (1986). *Social Foundations of Thought and Action*. Englewood Cliffs, NJ: Prentice-Hall.

Belova, N., and Zowada, C. (2020). Innovating higher education via game-based learning on misconceptions. *Educ. Sci.* 10 (9), 221. doi: 10.3390/educsci10090221

Bennacer, I. (2022). *Teaching analytics: support for the evaluation and assistance in the design of teaching through artificial intelligence*. (Doctoral dissertation). Le Mans Université, HAL. Available online at: https://theses.hal.science/2022LEMA1032 (Accessed June 16, 2025).

Brauer, S. (2019). *Digital open badge-driven learning: Competence-based professional development for vocational teachers*. (Doctoral dissertation). University of Lapland. doi: 10.13140/RG.2.2.21525.78565

Carrion Candel, E., De-la-Pena, C., and Chaves-Yuste, B. (2025). A gamified digital framework in higher education: impact on learning and motivation. *Turk. Online J. Dist. Educ.* 26 (3), 60–84. doi: 10.17718/tojde.1428690

Chan, S., and Lo, N. (2022). Teachers' and students' perception of gamification in online tertiary education classrooms during the pandemic. *SN Comput. Sci.* 3 (3), 215. doi: 10.1007/s42979-022-01117-w

Cosme-Jesús, J. M., Monteagudo-Fernández, J., Moreno-Vera, J.-R., and Sainz-Gómez, M. (2019). Effects of a gamification and flipped-classroom program for teachers in training on motivation and learning perception. *Educ. Sci.* 9 (4), 299. doi: 10.3390/educsci9040299

Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. New York, NY: Harper and Row.

Dahlan, F., Alias, N., and Shaharom, M. S. N. (2024). Gamification and game-based learning for vocational education and training: a systematic literature review. *Educ. Inf. Technol.* 29 (2), 1279–1317. doi: 10.1007/s10639-022-11548-w

De Sousa Mendes, D., and de Lima, M. R. (2021). Gamificación, "No tengo ni idea de lo que es": Un estudio en la formación inicial del profesorado de educación física. *Alteridad* 17 (1), 95–104. doi: 10.17163/alt.v17n1.2022.01

Deci, E. L., and Ryan, R. M. (1985). *Intrinsic Motivation and Self-determination in Human Behavior*. New York, NY: Plenum Press.

Desimone, L. M. (2009). Improving impact studies of teachers' professional development: toward better conceptualizations and measures. *Educ. Res.* 38 (3), 181–199. doi: 10.3102/0013189X08331140

Deterding, S., Sicart, M., Nacke, L., O'Hara, K., and Dixon, D. (2011). "Gamification. using game-design elements in non-gaming contexts," in *CHI '11 Extended Abstracts on Human Factors in Computing Systems*, eds. D. S. Tan, S. Amershi, B. Begole, W. A. Kellogg, and M. Tungare (New York, NY: ACM), 2425–2428.

Durukan, A. (2023). *The interplay between teacher knowledge and creativity: Insights from the extended reality assisted learning design process of pre-service science teachers*. (Doctoral dissertation), Middle East Technical University. doi: 10.13140/RG.2.2.12722.58568

Escobosa Morera, G., Carbonero-Sánchez, L., and Prat, M. (2024). Fitcoin race: una propuesta de gamificación para trabajar los hábitos saludables en la formación inicial del profesorado. *Retos* 51, 1234–1244. doi: 10.47197/retos.v51.98807

Fabre-Mitjans, N., Jiménez-Valverde, G., Guimerà-Ballesta, G., and Calafell-Subirà, G. (2025). Digital gamification to foster attitudes toward science in early childhood teacher education. *Appl. Sci.* 15 (11), 5961. doi: 10.3390/app15115961

Franco-Mariscal, A. J., Cebrián-Robles, D., and Rodríguez-Losada, N. (2023). Impact of a training programme on the e-rubric evaluation of gamification resources with pre-service secondary school science teachers. *Technol. Knowl. Learn.* 28 (2), 769–802. doi: 10.1007/s10758-021-09588-1

Guerrero Puerta, L. (2024). Exploring if gamification experiences make an impact on pre-service teachers' perceptions of future gamification use: a case report. *Societies* 14 (1), 11. doi: 10.3390/soc14010011

Guskey, T. R. (2002). Professional development and teacher change. *Teach. Teach.* 8 (3), 381–391. doi: 10.1080/135406002100000512

Güzel, S. (2023). Gamified lesson plan development with ELT pre-service teachers: a gamified Google Classroom experience. *J. Educ. Technol. Online Learn.* 6 (3), 544–564. doi: 10.31681/jetol.1158404

Hamari, J., Koivisto, J., and Sarsa, H. (2014). "Does gamification work? A literature review of empirical studies on gamification," in *Proceedings of the 47th Annual Hawaii International Conference on System Sciences*, (Maui, HI: IEEE), 3025–3034. doi: 10.1109/HICSS.2014.377

Hong, Q. N., Fàbregues, S., Bartlett, G., Boardman, F., Cargo, M., Dagenais, P., et al. (2018). The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Educ. Inf.* 34 (4), 285–291. doi: 10.3233/EFI-180221

Kapp, K. M. (2012). *The Gamification of Learning and Instruction: Game-Based Methods and Strategies for Training and Education*. San Francisco, CA: John Wiley and Sons.

Krishnan, S. D., Norman, H., and Md Yunus, M. (2021). Online gamified learning to enhance teachers' competencies using Classcraft. *Sustainability* 13 (19), 10817. doi: 10.3390/su131910817

Landers, R. N. (2014). Developing a theory of gamified learning: linking serious games and gamification of learning. *Simul. Gaming.* 45 (6), 752–768. doi: 10.1177/1046878114563660

León, A. M., Camacho-Lazarraga, P., Guerrero-Puerta, L., Guerrero, M. A., Aguilar-Parra, J. M., Colmenero-Ruiz, M. J., et al. (2022). Gamification in initial teacher training through a super-professor web application. *Educ. Sci.* 12 (1), 29. doi: 10.3390/educsci12010029

Liu, T., Oubibi, M., Zhou, Y., and Fute, A. (2023). Research on online teachers' training based on the gamification design: a survey analysis of primary and secondary school teachers. *Heliyon* 9 (4), e15053. doi: 10.1016/j.heliyon.2023.e15053

Lo, N. P., and Mok, B. C. (2019). "Gaming literacy and its pedagogical implications," in *Digital Humanities and New Ways of Teaching*, ed. A. W. Tso (Singapore: Springer), 133–154. doi: 10.1007/978-981-13-1277-9_8

Mahat, J., Alias, N., and Yusop, F. D. (2022). Systematic literature review on gamified professional training among employees. *Interact. Learn. Environ.* 32 (5), 2043–2064. doi: 10.1080/10494820.2022.2043910

Marín-Díaz, V., Sampedro-Requena, B. E., Muñoz-Gonzalez, J. M., and Jiménez-Fanjul, N. N. (2020). The possibilities of gamifying the mathematical curriculum in the early childhood education stage. *Math.* 8 (12), 2215. doi: 10.3390/math8122215

Martínez Sánchez, R. (2024). Transforming online education: the impact of gamification on teacher training in a university environment. *Metaverse Basic Appl. Res.* 2, 47. doi: 10.56294/mr202347

Mc Kevitt, C., Porcenaluk, S., and Connolly, C. (2025). Effective professional development and gamification enacting curriculum changes in critical mathematics education. *Educ. Sci.* 15 (7), 843. doi: 10.3390/educsci15070843

Mekler, E. D., Brühlmann, F., Tuch, A. N., and Opwis, K. (2017). Towards understanding the effects of individual gamification elements on intrinsic motivation and performance. *Comput. Human. Behav.* 71, 525–534. doi: 10.1016/j.chb.2015.08.048

Mishra, P., and Koehler, M. J. (2006). Technological pedagogical content knowledge: a framework for teacher knowledge. *Teach. Coll. Rec.* 108 (6), 1017–1054. doi: 10.1111/j.1467-9620.2006.00684.x

Nicholson, S. (2015). "A RECIPE for meaningful gamification," in *Gamification in Education and Business*, eds. T. Reiners, and L. Wood (Cham: Springer), 1–20. doi: 10.1007/978-3-319-10208-5_1

Okoli, C. (2015). A guide to conducting a standalone systematic literature review. *Commun. Assoc. Inf. Syst.* 37 (1), 43. doi: 10.17705/1CAIS.03743

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Br. Med. J.* 372, n71. doi: 10.1136/bmj.n71

Polyakova, O., and de Ros Cócera, L. (2022). Educational breakout and sustainable CLIL teacher training. *Psychol. Sci. Educ.* 27 (2), 96–107. doi: 10.17759/pse.2022270208

Rapp, A., Hopfgartner, F., Hamari, J., Linehan, C., and Cena, F. (2019). Strengthening gamification studies: current trends and future opportunities of gamification research. *Int. J. Hum. Comput. Stud.* 127, 1–6. doi: 10.1016/j.ijhcs.2018.11.007

Rethlefsen, M. L., Kirtley, S., Waffenschmidt, S., Ayala, A. P., Moher, D., Page, M. J., et al. (2021). PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews. *Syst. Rev.* 10 (1), 39. doi: 10.1186/s13643-020-01542-z

Rodrigues, L., Toda, A. M., Palomino, P. T., Oliveira, W., and Isotani, S. (2022). Gamification suffers from the novelty effect but benefits from the familiarization effect: findings from a longitudinal study. *Int. J. Educ. Technol. High. Educ.* 19, 13. doi: 10.1186/s41239-021-00314-6

Romero-Rodrigo, M., and López-Marí, M. (2021). Luces, sombras y retos del profesorado entorno a la gamificación apoyada en TIC: un estudio con maestros en formación. *Rev. Electron. Interuniv. Form. Profesorado* 24 (2), 167–179. doi: 10.6018/reifop.470991

Sailer, M., Hense, J. U., Mayr, S. K., and Mandl, H. (2017). How gamification motivates: an experimental study of the effects of specific game design elements on psychological need satisfaction. *Comput. Human. Behav.* 69, 371–380. doi: 10.1016/j.chb.2016.12.033

Seaborn, K., and Fels, D. I. (2015). Gamification in theory and action: a survey. *Int. J. Hum. Comput. Stud.* 74, 14–31. doi: 10.1016/j.ijhcs.2014.09.006

Skinner, B. F. (1938). *The Behavior of Organisms*. New York, NY: Appleton-Century-Crofts.

Stakhova, I., Kushnir, A., Franchuk, N., Kolesnik, K., Lyubchak, L., and Vatso, M. (2024). Enhancing the digital competence of prospective primary school teachers through utilizing Kahoot. *Int. Electron. J. Elem. Educ.* 16 (4), 467–478. doi: 10.26822/iejee.2024.346

Torrado Cespón, M., and Bárcena Toyos, P. (2025). An example of gamification for pre-service teachers in online higher education: methods, tools, and purpose. *Digit. Educ. Rev.* 46, 1–14. doi: 10.1344/der.2025.46.1-14

Vrcelj Božić, A., Hoić-Božić, N., Holenko Dlab, M., Stančin, K., and Jagušt, T. (2025). Gamification classification as a framework for analysing gamification elements in digital educational tools. *JCOMSS.* 21 (1), 34–42. doi: 10.24138/jcomss-2024-0129

Yáñez-Sepúlveda, R., Hinojosa-Torres, C., Cortés-Roco, G., and Zavala-Crichton, J. P. (2024). Aprendizaje basado en proyectos y gamificación como estrategias de aprendizaje en la formación de profesores de Educación Física. *Retos* 60, 1–11. doi: 10.47197/retos.v60.103422